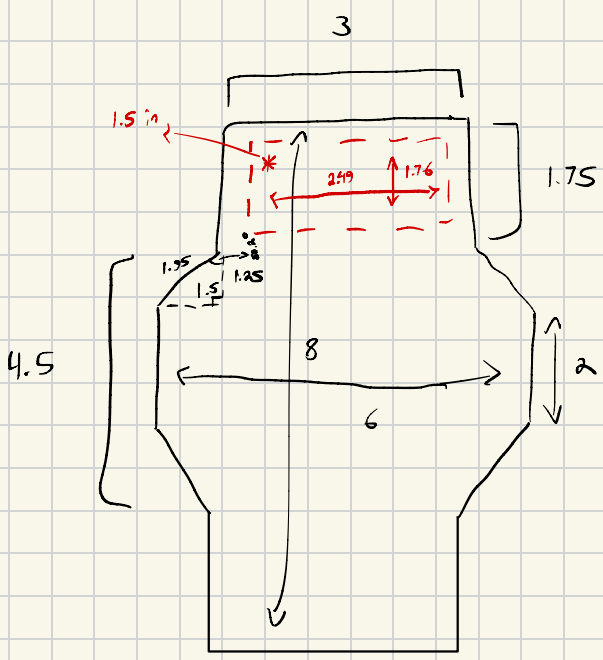
PLA

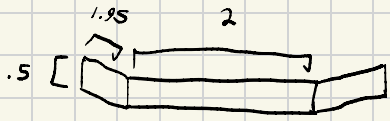
Density: 1.25 g/cm³ → .04516 lbf/in³

Board Weight

$pV = .04516 \cdot 11.615$

$= .525$ lbs





Wall Area = $2(.5) \cdot 2 + 1.95(.5) \cdot 4$

$= 5.9$ in²

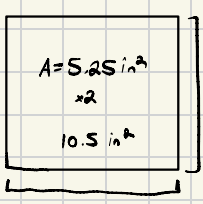
Wall Volume = $.885$ in³

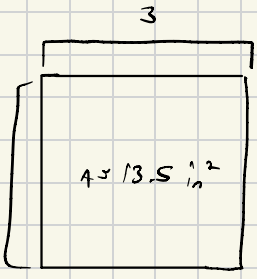
Board Volume

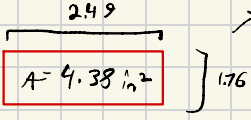
$.25(10.5 + 13.5 + 9.76) + 2.29$

$+ .885$

$= 11.615$ in³







4.38 in² · 1.5 in = 6.57 in³

Thickness: 1.5 in

Circular Cut Radius: .74 in

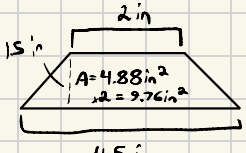
Circular Cut Volume: $\pi r^2 h$

$= \pi(.74)^2 2.49 = 4.28$ in³

Motor mount Volume:

$6.57 - 4.28$

$= 2.29$ in³



Part Weights: Battery: 360 g → .79 lbs

Servo: 37 g → .082 lbs

ESC: 39 g → .086 lbs

Motor: 168 g → .386 lbs

Reciever: 6 g → .013 lbs

Wheels: 138 g → .304 lbs

Axel Gear:

$\rho = .283 \text{ lb/in}^3$

$V = \pi R^2 h + \pi r^2 h - \pi r_6^2 h$

$= \pi\left(\frac{1.22}{2}\right)^2(.2) + \pi\left(\frac{.1}{2}\right)^2(.47-.2) - \pi\left(\frac{.24}{2}\right)^2(.47)$

$= .47 \text{ in}^3$

$W = .47 \cdot .283 = .13 \text{ lbs}$

Motor Gear:

$\text{mod} = \frac{d}{z}$ $\quad d = \text{mod}\, z = .5(12) = 6\text{mm}$ $\quad h \approx 7\text{mm}$

$\pi R^2 h - \pi r_6 h$ $\quad \rho \approx .282$

$\pi(6)^2(7) - \pi(3)^2(6)$

$= 622 \text{ mm}^3 = .037$ $\quad W = .037 \cdot .282 = .010 \text{ lb}$

Driveshaft:

$\pi R^2 h = \pi(.098)^2(3.94) = .118 \text{ in}^3$

$\rho = .282$ $\quad W = .118 \cdot .282 = .03 \text{ lb}$

Part Total: 1.834 lbs

Car Weight: 2.36 lbs $\quad$ Car Mass: 1.07 kg

Top Speed:

Motor Max ω = 23,400 RPM

Add Gearing: $\frac{23{,}400}{4}$ = 5850 RPM @ Wheels

Convert: $5850 \cdot 2\pi = 36756.63$ rad/m = 2205398.043 rad/h

Linearize: $V = \omega r = 2205398.043 \text{ rad/h} \cdot .0508 \text{ m} = 112034.2206$ m/h

Convert to MPH/Account for Efficiency $= \frac{112034 \cdot .25}{1609} =$ **17.4 MPH**

Maximum Acceleration:

Limiting factor?

ESC: 60 A cont. ✓ Motor Stall: 70 A

Estimate Torque Constant (Stall/Max Efficiency)

$K_t = \frac{\tau}{I} = \frac{196E-3}{70} = .0028$ $\quad \frac{30.6}{13} = .00235$ $\quad K_t = \frac{.0028 + .00235}{2}$

$= .0026$

Estimate Torque at 60 A

$$\tau = K_t \cdot I = .0026 \cdot 60 = .15 \text{ N·m}$$

Tractive Effort:

* Assume wheel radius 2 in ≈ .0508 m and gear ratio 4:1, ignore drag

$$F = \frac{\tau \cdot \eta_e \cdot N_{tp}}{r} = \frac{.15 \cdot .3 \cdot 4}{.0508} = 3.54 \text{ N}$$

Motor Speed @ 60 A 5000 RPM

Convert to Wheel Speed:

$$\frac{5000 \cdot 2\pi \cdot 60 \cdot .0508 \cdot .28}{4.1609} = 4.17 \text{ MPH}$$

Rolling Resistance:

Radial Tires: $f_r = \left(.0041 + .000041V\right) C_h$ ← Road Surface Coefficient

$C_h = 1.5$ on concrete

* 60 A draw

@ 4.2 mph (1.86 m/s)

$$f_r = \left(.0041 + .000041(1.86)\right) 1.5 = .0062$$

$R_x = F_r W$

$$R_x = .0062 (8.39) = .053 \text{ N}$$

Acceleration

$\Sigma F = ma$ $\quad a = \frac{F - F_R}{m} = \frac{3.54 - .053}{1.07} = 3.26 \text{ m/s}^2$